Does Seeing the Search Result Make Tool Utility Predictable?

A diagnostic follow-up to *To Call or Not to Call* (Wu et al., 2026)

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1. Question

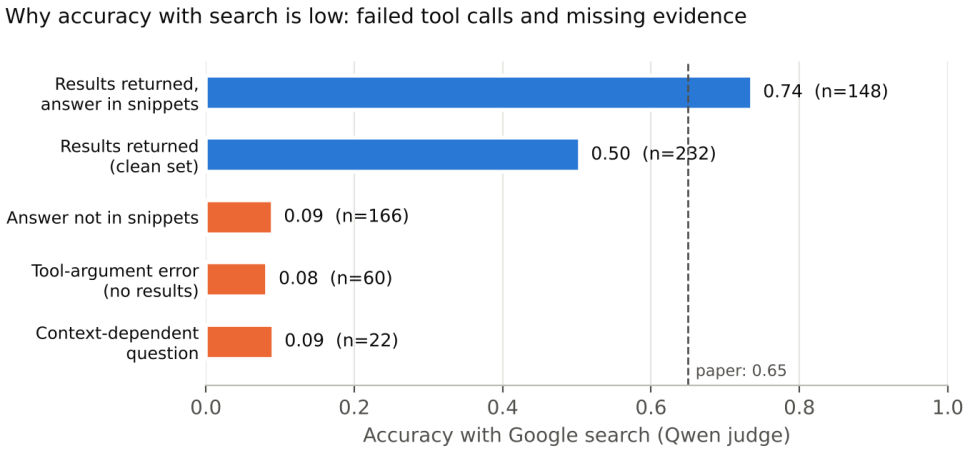
Wu et al. find that a hidden-state probe predicts whether an LLM **needs** a tool well (LNE) but predicts whether the tool will **help** (LUE) less well, and attribute this to utility depending on the tool output, which the model has not yet seen. We test two alternatives: (a) the utility labels are noisy (each comes from one no-tool run, one tool run and one LLM-judge call); (b) the information-bottleneck explanation holds, in which case a probe that sees the retrieved results should predict utility much better.

2. Setup

Llama-3.2-3B-Instruct (ungated copy of the same weights), the authors' harness unmodified (base loop, vLLM, temperature 0); BFCL web-search split (314 questions); conditions no-tool vs. always-search with Google (SerpAPI top-5 snippets, Oct 2026). Judge: the paper's exact BFCL judge prompt on Qwen2.5-7B-Instruct (4-bit, temperature 1.0) instead of GPT-4o; every answer judged 3 times. Labels: need = no-tool answer wrong; utility = tool score > no-tool score. Probes: the paper's MLP grid on final-layer last-token states. **Pre** = no-tool prompt (as LUE_x); **post** = final prompt containing the retrieved snippets. Nested 5-fold CV (hyperparameters chosen inside each training fold), identical outer folds for pre and post, seed 42. Bootstrap 95% CIs over questions; paired bootstrap for the pre-vs-post difference.

3. Data-quality checks

Always-search accuracy on all 314 questions was 0.40 (paper 0.65), so we traced the cause before analysing probes. Time-sensitive questions are **not** the cause (0.41 on "current/latest" vs. 0.41 on stable questions). In **60/314 runs (19%) the model called the search tool with a wrong argument name**, so the harness returned an error string instead of results (accuracy 0.08); 22 questions are context-dependent ("Where is his hometown?", accuracy 0.09); 2 hit an API quota. Error strings in the post prompt would be an easy cue for "search did not help", so we analyse the **clean set** (232 questions; 229 with hidden states) as primary and report the full set alongside. On the clean set accuracy is 0.12 without and 0.50 with Google; the gold answer appears in the retrieved snippets for only 47% of questions (accuracy 0.74 when it does, 0.09 when not), pointing to retrieval quality.

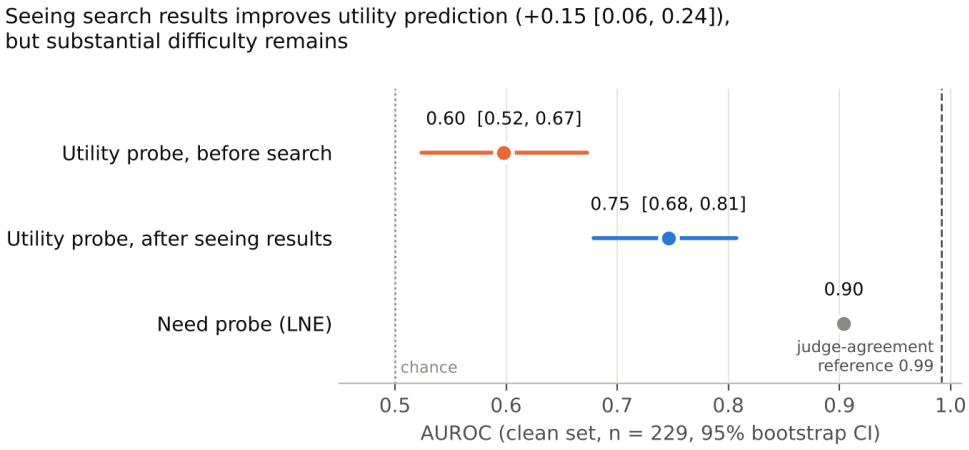


4. Results

Q1, label reliability. Utility labels re-judged three times: Fleiss κ 0.97 [0.94, 0.99] (clean set); per-question flip rate 0.03 [0.01, 0.05] on the full set. Repeated judgments are highly consistent, so repeat-to-repeat disagreement alone is unlikely to explain the gap. Consistency is not correctness: an independent second model (Claude) labelled a stratified sample of 50 answers blind to the judge and agreed 90% of the time (Cohen's κ 0.80); 4 of 5 disagreements were the judge marking a correct answer wrong.

Q2, does seeing the evidence help?

Probe (clean set, n = 229)	**AUROC [95% CI]**
Need probe (LNE)	0.90
Utility probe, pre-search	0.60 [0.52, 0.67]
Utility probe, post-search (sees results)	0.75 [0.68, 0.81]
Difference post - pre (paired bootstrap)	**+0.15 [0.06, 0.24]**
Full uncleaned set (n = 308), for comparison	0.68 -> 0.82, +0.14 [0.07, 0.21]



5. What this means

Access to the tool output carries real signal: utility prediction improves by +0.15 AUROC, supporting the paper's explanation in part. But even with the evidence in context, substantial predictive difficulty remains (0.75, against a judge-agreement reference of 0.99 that reflects label consistency only, not a theoretical maximum). Competing explanations for the residual, which we have not separated, are retrieval quality (evidence often lacks the answer), whether the model actually uses the evidence, answer-generation errors, and limits of a last-token probe. The post-search probe is a diagnostic, not a deployable pre-call controller.

6. Differences from the original setup and limitations

Qwen judge instead of GPT-4o; searches run in Oct 2026; temperature 0 (repo default 1.0); one model, one engine, one task; the clean-set filter is post hoc (full-set result reported alongside); the correctness check uses a second model, not human annotators. **Next:** condition the post-search probe on whether the evidence contains the answer, add a second search engine to measure search-side label noise, and repeat on larger models and the paper's other tasks.

References: Wu et al. 2026, *To Call or Not to Call*, arXiv:2605.00737. Asai et al. 2024, Self-RAG. Yan et al. 2024, Corrective RAG.